# Episode 7: The Numbers Deck



Programming with Python · Tutorial 07

The investor liked last week's numbers. Now she wants a **one-page metrics deck**: totals, a per-zone breakdown, the busiest day, the strongest zone. Tobi's counter-proposal, a 40-tab spreadsheet with a formula pointing at a workbook he can't find, is gently disqualified. So this week you learn the tool that turns a pile of orders into metrics without a single hand-written loop: **NumPy**. You'll build arrays, multiply a whole price list in one expression, use boolean **masks** to count and filter, and total a 2-D grid by row or by column with `axis`.

## Work on the notebook

Open in browser 


**Open in browser (recommended):** runs entirely on your machine, in this tab: no account, no installation, and after it loads no internet is needed.

> ! **How your work is saved (read this once)**
> Your progress lives **in this browser tab**. If you **reload** the page (Cmd/Ctrl + R) your work is still there. But if you **close the tab and open the link again later, you start from a clean notebook**. There is no cross-device sync in the browser, and clearing browser data or private mode also wipes it.
>
> So: **download your `.py` before you leave**: press **Cmd/Ctrl+S**, then menu → Download → *Download Python code*. Without the save, the file is empty or misses your last edits. That download is the *only* guaranteed copy. Handing in files works exactly like this in the checkpoints, so you get to practice the motion every week. Note: labs are not graded and nothing is handed in. The downloaded `.py` is your backup. You can't upload it back into the browser editor.

**Open in molab:** marimo's free cloud (account required). Your copy saves to your account and reopens on **any device**. Choose this if you know you'll switch computers or want to be certain nothing is lost. Optional: nothing graded ever requires it.

## The install from this session

In Session VII we also install **uv**, **Zed** and **Mistral Vibe** together in class, because from Session VIII the labs run on your own laptop. Missed it? Do it at home before Session VIII: first the uv guide from the top through *Adding packages*, then the six steps under *AI inside Zed* on the AI Tools page. The last step is the check we did in class.

# Bibliography